

1.linkedlist

```
public class Main {  
    static Node head = null;  
  
    // Insert at Beginning  
    static void insertFirst(int data) {  
        Node newNode = new Node(data);  
        newNode.next = head;  
        head = newNode;  
    }  
  
    // Insert at End  
    static void insertLast(int data) {  
        Node newNode = new Node(data);  
        if (head == null) {  
            head = newNode;  
            return;  
        }  
        Node temp = head;  
        while (temp.next != null) {  
            temp = temp.next;  
        }  
        temp.next = newNode;  
    }  
  
    // Insert at Position (1-based)  
    static void insertPosition(int pos, int data) {  
        if (pos <= 0) {  
            System.out.println("Invalid Position");  
            return;  
        }  
        if (pos == 1) {
```

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```
        insertFirst(data);
        return;
    }
    Node newNode = new Node(data);
    Node temp = head;
    for (int i = 1; i < pos - 1; i++) {
        if (temp == null) {
            System.out.println("Invalid Position");
            return;
        }
        temp = temp.next;
    }
    if (temp == null) {
        System.out.println("Invalid Position");
        return;
    }
    newNode.next = temp.next;
    temp.next = newNode;
}
```

// Delete First

```
static void deleteFirst() {
    if (head == null) {
        System.out.println("List is Empty");
        return;
    }
    head = head.next;
}
```

// Delete Last

```
static void deleteLast() {
```

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```
if (head == null) {
    System.out.println("List is Empty");
    return;
}
if (head.next == null) {
    head = null;
    return;
}
Node temp = head;
while (temp.next.next != null) {
    temp = temp.next;
}
temp.next = null;
}

// Delete at Position (1-based)
static void deletePosition(int pos) {
    if (head == null) {
        System.out.println("List is Empty");
        return;
    }
    if (pos <= 0) {
        System.out.println("Invalid Position");
        return;
    }
    if (pos == 1) {
        head = head.next;
        return;
    }
    Node temp = head;
    for (int i = 1; i < pos - 1; i++) {
```

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```
        if (temp == null) {
            System.out.println("Invalid Position");
            return;
        }
        temp = temp.next;
    }
    if (temp == null || temp.next == null) {
        System.out.println("Invalid Position");
        return;
    }
    temp.next = temp.next.next;
}
```

// Display

```
static void display() {
    if (head == null) {
        System.out.println("List is Empty");
        return;
    }
    Node temp = head;
    while (temp != null) {
        System.out.print(temp.data + " -> ");
        temp = temp.next;
    }
    System.out.println("null");
}
```

```
public static void main(String[] args) {
    System.out.println("Insert Operations");
    insertFirst(20);
    insertFirst(10);
}
```

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```
insertLast(30);
```

```
insertPosition(3, 25);
```

```
display();
```

```
System.out.println("\nDelete First");
```

```
deleteFirst();
```

```
display();
```

```
System.out.println("\nDelete Last");
```

```
deleteLast();
```

```
display();
```

```
System.out.println("\nDelete Position");
```

```
deletePosition(2);
```

```
display();
```

```
}
```

```
}
```

```
class Node {
```

```
    int data;
```

```
    Node next;
```

```
    Node(int data) {
```

```
        this.data = data;
```

```
        this.next = null;
```

```
}
```

```
}
```